

Module 24.04: Overfitting (High Variance & Memorizing Noise)

1. Theoretical Definition of Overfitting

Overfitting occurs when a machine learning model fits too closely or exclusively to a specific dataset, learning the empirical random noise, statistical fluctuations, and idiosyncrasies rather than the true underlying data-generating distribution $P(X, Y)$.

Target Ground Truth: $y = f(x) + \epsilon$ (where $\epsilon \sim N(0, \sigma^2)$ is irreducible noise)

Overfitted Model: Learns $f(x) + \epsilon$ perfectly!

Result:

* Training Error $\rightarrow 0.0$ (Near zero loss, $R^2 \sim 1.0$)

* Test/Val Error $\rightarrow$ Very Large (Catastrophic failure on unseen real-world data)

2. The Bias-Variance Decomposition Perspective

For any regression model $\hat{f}(x)$, the expected generalization error on an unseen test point x^* decomposes strictly into three orthogonal components:

$$\mathbb{E} \left[(y - \hat{f}(x^*))^2 \right] = \text{Bias}^2 \left[\hat{f}(x^*) \right] + \text{Var} \left[\hat{f}(x^*) \right] + \sigma^2$$

Where:

- $\text{Bias} \left[\hat{f}(x^*) \right] = \mathbb{E} \left[\hat{f}(x^*) \right] - f(x^*)$: Error introduced by simplifying assumptions.
- $\text{Var} \left[\hat{f}(x^*) \right] = \mathbb{E} \left[(\hat{f}(x^*) - \mathbb{E}[\hat{f}(x^*)])^2 \right]$: The sensitivity of the model to the specific training dataset drawn.
- σ^2 : Irreducible noise variance inherent in the physical process.

Mathematical Anatomy of High Variance:

In an overfitted model:

1. **Bias** ≈ 0 : The model has excessive capacity to bend through every training point.
2. **Variance** $\rightarrow$ **Extremely High**: Training on dataset $\mathcal{D}_1$ vs $\mathcal{D}_2$ produces radically disparate weight parameters.
3. The model exhibits explosive sensitivity to infinitesimal shifts in inputs.

3. Causes and Symptoms of Overfitting

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| Root Causes of Overfitting
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| 1. Excessive Model Complexity: High polynomial degrees (e.g., degree 15 on 20 p
| 2. Insufficient Training Samples: Sample size m is too small relative to featur
| 3. Noisy / Corrupted Data: Training set contaminated with outliers and mislabel
| 4. Unconstrained Weights: Model parameters w_j grow into massive magnitudes.
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Signature Diagnostic Sign:

A large and expanding **generalization gap**:

$$\text{Gap} = \mathcal{L}(\text{val}) - \mathcal{L}(\text{train}) \gg 0$$

As model capacity increases, $\mathcal{L}(\text{train})$ *monotonically descends*, while $\mathcal{L}(\text{val})$ initially drops but then violently explodes upwards.